

EXPLORATION BY RANDOM NETWORK DISTILLATION

Poster Presentation – Group #7

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Problem/Objective

Something

Methodology

Intrinsic Reward

In sparse-reward environments, RL agents may recieve little to no feedback from environment.

To encourage exploration, an intrinsic reward component is added to the reward signal.

Total reward becomes the sum of extrinsic (e_t) and intrinsic reward (i_t):

$$r_t = e_t + i_t$$

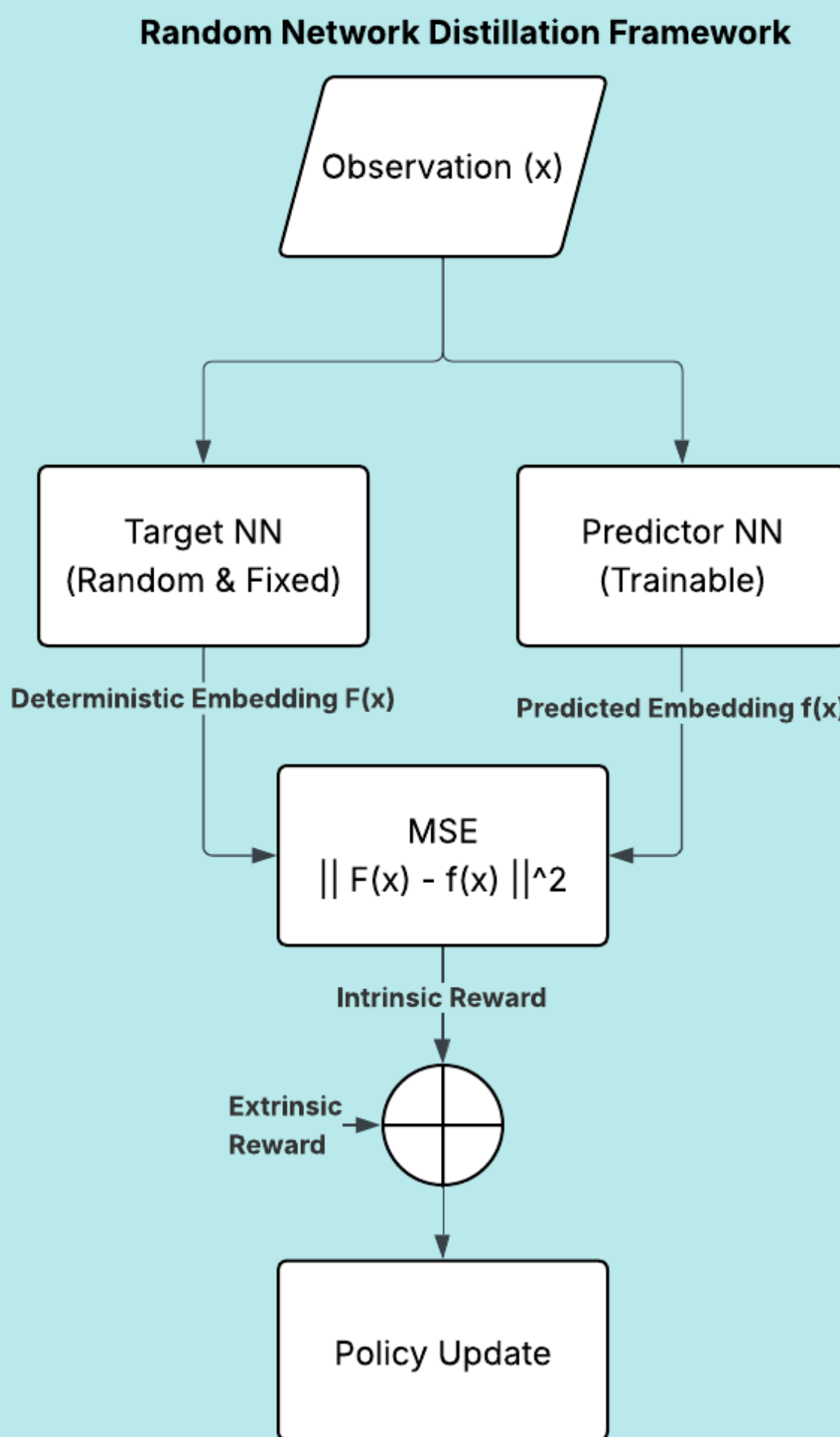
i_t is large for unfamiliar states and decreases as states become familiar.

Random Network Distillation (RND)

Intrinsic reward is computed as the MSE between two neural networks:

- **Target Network:** Randomly initialized & fixed.
- **Predictor Network:** Tries to imitate the target.

When encountering a new state, the predictor network produces a large error $\rightarrow$ Strong intrinsic reward. As a state is revisited, the predictor network learns them, the MSE decreases, and the reward decreases.



Results

Something

Limitiations

Something

Why It Matters?

Something

Conclusion

Something

Key Sources & Acknowledgements